BURUGUNTE DEVELOPERS

# MUSCLE CONTRACTION TRAINER

ReGain: play games with your muscle signal

BioAmp EXG Pill · Arduino · Python

01 · PROBLEM STATEMENT

# PATIENTS CAN'T SEE THEIR MUSCLE WORK

Muscle rehab relies on isometric contractions: tightening a muscle without moving the joint. The effort is invisible and repetitive, so patients lose motivation and stop attending.

| | |
|---|---|
| **No feedback** | Neither patient nor therapist can easily tell if the muscle switches on, or how hard. |
| **Costly to measure** | Commercial EMG biofeedback units cost about ₹20,000 to ₹59,000. |
| **Low motivation** | Repeating plain holds is dull, which hurts retention in a therapy course. |

02 · OUR SOLUTION

# A ₹3,100 EMG TRAINER THAT TURNS EFFORT INTO A GAME

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|---|---|
| **See it** | A live bar shows effort from 0 to 100% of the patient's own maximum. |
| **Play it** | Games reward holding the right effort level, so repetition feels like play. |
| **Track it** | A fatigue monitor shows when the muscle starts to tire during a hold. |

**EMG (electromyography)** means measuring the tiny electrical signals a muscle makes when it contracts.

03 · HOW IT WORKS

# FROM MUSCLE TO GAME

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|---|---|
| **Muscle** | Active fibres make tiny voltages on the skin. **Electrodes** are sticky pads that pick them up. |
| **BioAmp EXG Pill** | Compares two electrodes and amplifies the **difference**, so noise common to both (like mains hum) cancels. |
| **Arduino** | Its **ADC** (analog-to-digital converter) turns voltage into a number from 0 to 1023, 500 times a second. |
| **Python** | Filters the numbers, measures signal size, and rescales to your own range. |
| **Game** | Receives one number, **activation**, from 0 (relaxed) to 1 (your max). |

04 · CLEANING THE SIGNAL

# FOUR STEPS BEFORE THE GAME SEES IT

| | |
|---|---|
| **1. Subtract 512** | The board can't output negative voltage, so it parks the signal's zero at half the ADC range (512 of 1023). Subtracting 512 restores true positive and negative swings. |
| **2. 50 Hz notch** | **Hz** means cycles per second. Mains power hums at 50 Hz in India. A notch filter cuts only a thin band around that frequency. |
| **3. 20 Hz high-pass** | Blocks slow drift below 20 Hz from moving electrodes and sweat. Useful EMG energy sits roughly between 20 and 250 Hz. |
| **4. RMS over 200 ms** | **Root mean square:** square each sample, average the last 100, take the square root. Squaring stops positive and negative swings cancelling to zero, so the result tracks how big the swings are. |

05 · CALIBRATION AND ACTIVATION

# SCALING THE SIGNAL TO YOU

Raw numbers differ by person, muscle and day. So we measure your own floor and ceiling of the **envelope** (the smoothed RMS from the last step).

**Rest = 6.4:** relax for 3 s, skip the first 20%, take the average.

**Max = 200.4:** squeeze hard for 3 s and take the **95th percentile**, the value 95% of readings sit below, so one spike can't set an impossible max.

## ACTIVATION = (ENVELOPE − REST) ÷ (MAX − REST)

Clamped so it never goes below 0 or above 1.

| Envelope | Activation |
|---|---|
| 6.4 | 0.00 |
| 55 | 0.25 |
| 103 | 0.50 |
| 152 | 0.75 |
| 200+ | 1.00 |

06 · SPOTTING FATIGUE

# TIRED MUSCLE SHIFTS TO LOWER FREQUENCY

Tired fibres conduct pulses more slowly, so signal power moves to lower frequencies while its size can stay the same. That's why we watch frequency, not amplitude.

**Power spectrum:** how much power sits at each frequency, like equaliser bars. **MDF (median frequency):** the frequency that splits total power in half. We compute it from 0.5 s windows, four times a second.

| Frequency | 40 Hz | 60 Hz | 80 Hz | 100 Hz | 120 Hz | MDF |
|---|---|---|---|---|---|---|
| Fresh: power | 1 | 3 | 4 | 3 | 1 | 70 Hz |
| Tired: power | 1 | 4 | 4 | 2 | 1 | 65 Hz |

Example: total power is 12, so half is 6. Counting up from 40 Hz, 6 is reached at 70 Hz when fresh and at 65 Hz when tired.

07 · FATIGUE SCORE

# FROM MDF TO A FATIGUE PERCENTAGE

| | |
|---|---|
| **Gate** | Only windows above 10% activation count. A relaxed muscle gives a meaningless spectrum. |
| **Reference** | Median MDF from 0.5 s to 3 s of the hold is the fresh-muscle baseline. |
| **Smooth** | Rolling median of the last 5 values. The median is the middle value, so it ignores spikes. |

## FATIGUE % = (REFERENCE − CURRENT MDF) ÷ REFERENCE × 100

Example: reference 90 Hz, now 79 Hz gives about 12%. **Onset** is flagged when smoothed MDF stays below 88% of the reference for 3 windows in a row.

08 · THE GAMES

# TWO GAMES, ONE ARCADE

| | |
|---|---|
| **ReGain Flap** | Fly a bird through pipes by holding your muscle effort inside each gap. |
| **ReGain Climb** | Drive a car up hills. Hit the sweet spot, then relax to refuel. |

Launched from one retro arcade window. Each game calibrates separately. Demo mode uses the keyboard, so it runs without hardware. Not a medical device: use it under a physiotherapist's plan.

09 · COST

# WHAT DOES IT COST?

## ₹3,100

one-time hardware for the physiotherapist

## ₹16 TO ₹93

extra cost per session

## ₹190 TO ₹1,100

extra for the patient over a 12-session course

**Break-even:** it pays for itself if it improves retention, or shortens a course, by about 2 to 10%.

**₹3,100**
one-time hardware for the physiotherapist

**₹16 – 93**
extra cost per session

**₹190 – 1,100**
extra cost to the patient over a 12-session course

### CHART 1: UPFRONT COST VERSUS COMMERCIAL EMG BIOFEEDBACK




### CHART 2: EXTRA COST PER SESSION, BY SOURCE




**Break-even:** the tool pays for itself if it improves patient retention, or shortens a course of therapy, by about 2–10%.